

Seeing Materials in Context: An Immersive AR/XR Demo for Exploring the Interplay of Shape, Material, Colour, and Light

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Figure 1: Conceptual overview of the proposed XR-based interactive visualisation system. Users explore object appearance by manipulating shape, colour, material, and lighting through mid-air gestures, enabling real-time perceptual comparison without professional rendering tools. The illustration is AI-generated [14], based on a stylised human figure originally designed by the authors, and serves as a conceptual and indicative illustration rather than a depiction of the final user interface.

Abstract

Design decisions about object shape, material, and colour are inherently perceptual and often rely on designers' empirical judgement, as many digital tools fragment these attributes into parameter-centric workflows that hinder reasoning about appearance in context. We present an immersive AR/XR demonstration that enables integrated exploration of the interplay between shape, colour, material, and lighting. To support immersive use in XR, where mouse-and-keyboard input is unavailable, the system employs mid-air gestures for direct, embodied interaction. Illumination is represented through a small set of perceptually meaningful lighting contexts rather than low-level parameters, enabling real-time manipulation and rapid comparison across conditions. A decoupled interaction

architecture stabilises perceptual exploration despite imprecise gestural input, translating established findings from visual material perception into a hands-on tool for early-stage design exploration.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Mixed / augmented reality**; **Gestural input**.

Keywords

Perceptual Exploration, Design Exploration, Lighting Design, Embodied Interaction, Material Perception

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1 Introduction

Design decisions regarding shape, material, and colour are inherently perceptual, depending on subtle interactions between geometry, surface properties, and illumination. As noted by Schön, design is a reflective conversation with materials [16], yet digital workflows often disrupt this conversation by separating these factors. A material that appears appropriate under studio lighting may look overly glossy in an outdoor environment; a colour balanced in neutrality may become unstable when illumination changes.

Despite this, many design tools rely on parameter-centric interfaces (e.g., numerical sliders) that require designers to translate perceptual judgments into abstract values. Lighting is often treated as a secondary rendering concern rather than an active design variable. This separation hinders "reflection-in-action," forcing a disconnect between the designer's intent and the digital representation [10].

Rather than exposing low-level lighting parameters (e.g., ambient or spot), the system represents illumination through a small, curated set of perceptually meaningful lighting contexts (light maps). These contexts act as perceptual primitives that allow designers to reason about appearance directly, without translating judgments into abstract numerical controls.

We propose a design environment that foregrounds perception, treating lighting as an integral part of appearance exploration. Our objective is not to improve recognition performance, but to support *perception-driven design exploration*. Specifically, rapid perceptual comparisons could enable designers to assess how the same object behaves across different lighting contexts and to evaluate the stability of appearance under change.

To guide the system design, we articulate three design goals: (1) Support perceptual comparison across lighting contexts, enabling reasoning about appearance robustness rather than optimisation for a single condition; (2) Reduce cognitive interruption introduced by parameter-centric interfaces, allowing designers to focus on perceptual outcomes rather than numerical control; (3) Enable continuous, exploratory design behaviour through embodied interaction, framing design as an iterative perceptual dialogue rather than a sequence of discrete commitments.

To achieve this, we present a lightweight, non-intrusive system designed for walk-up-and-use exploration. It supports continuous adjustment of visual attributes across multiple lighting contexts via mid-air gestures. This interactivity demonstration invites attendees to experience "reflection-in-action" firsthand, manipulating shape, material, and light without the barrier of wearable devices.

2 Related Work

2.1 Appearance and Light-Material Interplay

A large body of vision science and perception research has examined how object appearance emerges from the interaction of lighting, shape, and material properties. Studies on visual material perception show that perceived surface qualities depend not only on intrinsic material attributes, but also on illumination structure, viewing conditions, and geometric context (e.g., [1, 9, 12, 15]). Collectively, this work emphasises that appearance is relational, arising from the interplay between light, material, and shape rather than from isolated parameters.

Within this literature, prior studies by the authors and collaborators have systematically investigated how lighting direction, surface geometry, and material characteristics influence perceived appearance under controlled conditions (e.g., [18–22]). These studies primarily relied on static stimuli or offline rendering pipelines to isolate perceptual factors and derive analytical models of light-material interaction.

While this research provides a strong perceptual foundation, its experimental methodologies differ fundamentally from contemporary design workflows. Professional appearance-editing tools typically expose low-level parameters for lighting, material, and geometry, requiring designers to translate perceptual judgments into abstract numerical controls. As a result, appearance evaluation is often decoupled from perceptual reasoning in context, and lighting is treated as a secondary rendering concern rather than an active design variable. This gap motivates the need for tools that embed perceptual knowledge directly into interactive appearance exploration.

2.2 Mid-air and embodied interaction

Our work bridges digital appearance editing and embodied interaction. Traditional editing tools rely on parameter-centric interfaces that decouple designers from perceptual outcomes, disrupting reflective engagement with materials [7, 16]. In contrast, embodied interaction frameworks treat action, perception, and meaning as inseparable [8].

Mid-air gestures have been widely explored as intuitive input for spatial and immersive systems since the 2010s (e.g. [2, 11]). In AR/XR environments, gesture-based interaction is often a practical necessity, as mouse-and-keyboard input is unavailable or incompatible with immersive use [3]. However, such interaction commonly suffers from limited precision and the "Midas Touch" problem, making it ill-suited to continuous parameter control.

Rather than mapping gestures directly to low-level parameters, our approach adopts a situated design perspective in which embodied input supports exploratory and reversible interaction. Using robust tracking (e.g., MediaPipe [17]) within a decoupled intent architecture, mid-air gestures are interpreted at the level of design intent, enabling continuous perceptual exploration with stability and robustness. This positions gesture input not as a replacement for traditional controls, but as an enabling modality for immersive, perception-driven design exploration.

2.3 Design exploration & reflection-in-action

From a design research perspective, appearance exploration is not merely a process of parameter optimisation, but a reflective practice in which designers iteratively reason about perceptual outcomes through action and feedback. Schön's notion of reflection-in-action characterises design as a continuous dialogue between intention, action, and perception [16]. Complementary perspectives in design research emphasise that designers rely on experiential knowledge, sketching, and iterative making to explore and refine ideas rather than to specify solutions upfront [4, 5].

However, many digital design tools interrupt this reflective loop by fragmenting perceptual factors and prioritising precise specification over exploratory interaction. Norman argues that such

systems often force designers to adapt their thinking to the tool, rather than allowing tools to support natural perceptual reasoning and sense-making [13]. This is particularly problematic when perceptual dependencies that are central to the design problem, such as the interaction between lighting, material, and shape.

The proposed CHI demo builds directly on established perceptual findings by translating them from laboratory-based analysis into an interactive, embodied exploration environment. Unlike prior presentations of perceptual research that relied on controlled stimuli or post hoc analysis, this demo enables real-time, gesture-based manipulation of shape, colour, material roughness, exposure, and lighting contexts. By foregrounding exploratory interaction over parameter specification, the system reframes perceptual knowledge as a tool for sense-making and creative exploration, aligning with CHI's emphasis on human-centred design, perception, and creativity.

3 Interaction and Experience Design

3.1 Design Rationale and System Overview

In the current implementation, visual feedback is presented on a conventional display using a spatial interaction metaphor. While the system is XR-inspired and XR-ready, it does not require head-mounted displays, supporting walk-up-and-use deployment on the demo floor.

As shown in Fig. 2, the system adopts a decoupled interaction architecture designed to support perceptual exploration rather than direct parameter control. Raw hand-tracking data is interpreted at the level of design intent (e.g., “adjust material smoothness” or “switch lighting context”), separating noisy, continuous input from the rendering parameters that determine visual output. This decoupling allows imprecise mid-air gestures to produce stable and meaningful perceptual changes, preserving a sense of direct manipulation without requiring users to manage low-level controls.

Importantly, gesture signals are not expressed as engine-specific commands or low-level parameter bindings, but as semantically meaningful design intents. This intermediate abstraction decouples input modality from rendering logic, allowing the perceptual environment to function as an intent consumer rather than a direct input handler.

For sensing and decoupling, a standard webcam is used to extract 21 hand landmarks. We propose a custom intent-decoupling layer translates raw coordinates into abstract signals and incorporates depth filtering to suppress background crowds as well as lighting normalisation to ensure robust tracking under uneven convention-centre illumination.

For rendering and performance, Unity maps intent signals to visual attributes (lighting, shape, colour, and material). The system runs stable on consumer hardware with low latency, ensuring that interaction remains responsive and supports a continuous reflective design process.

3.2 Gesture for Perceptual Exploration

The system employs a minimal, asymmetric gesture vocabulary designed to support safe and continuous perceptual exploration through bimanual interaction (Fig. 3). Rather than maximising expressiveness, the gesture set is intentionally constrained so that

navigation, continuous adjustment, confirmation, and cancellation are clearly separated in both form and function. This structure allows users to explore appearance changes fluidly while retaining explicit control over commitment.

Interaction responsibilities are distributed across the two hands to reduce cognitive load and prevent accidental input. The non-dominant hand supports sustained, reversible manipulation of perceptual attributes such as colour saturation or material smoothness, while the dominant hand is reserved for discrete actions that signal intent, including confirmation and context switching. Confirmation is implemented as a rhythmic, time-bound gesture rather than a static pose to mitigate the “Midas Touch” problem, with a complementary cancellation gesture allowing users to halt or undo actions when needed.

Together, this gesture vocabulary encodes a clear distinction between exploration and commitment, supporting reflection-in-action while remaining robust in the noisy conditions of a public demo environment.

3.3 The 3-Minute Walk-Up-and-Use Workflow

Attendees engage in a structured exploration loop that follows the logical progression of perceptual design (Fig. 4).

Light Map Selection (Context). Upon entering the tracking zone, the user activates the system with an open palm and uses the *Dynamic Page-Flip* gesture to browse perceptual lighting contexts, selecting environments such as a bright glacier or a shadowed forest.

Shape Selection. With the lighting context set, the user uses the same *Dynamic Page-Flip* gesture to cycle through and select a geometric shape as the design canvas.

Chromatic Definition (Colour Picker). The user then employs the *Pinch-Based Sustained Grab* (left hand) to navigate the HSV colour space, for example selecting a vibrant colour that contrasts with the Glacier environment.

Material Selection and Roughness Adjustment. Material properties are adjusted by switching the left-hand interaction mode, allowing continuous roughness control and transitions from matte ceramic to polished metal.

Final Lighting Adjustment. The user uses the right hand to rotate the lighting orientation and adjust exposure while holding a material state, enabling perceptual stress testing under different illumination conditions.

4 Discussion

This work reframes appearance design from parameter tuning toward perceptual exploration. By constraining illumination to a finite set of perceptually distinct lighting contexts, the system prioritises comparative reasoning and interpretability over exhaustive parameter freedom, which is particularly suitable for early-stage design. The workflow enables direct perceptual comparison across contexts, supporting reasoning about appearance stability and variation under change.

We further distinguish between lighting as contextual framing and lighting as perceptual validation. While HDR environments establish an initial perceptual hypothesis grounded in plausible usage scenarios, subsequent orientation and exposure adjustments

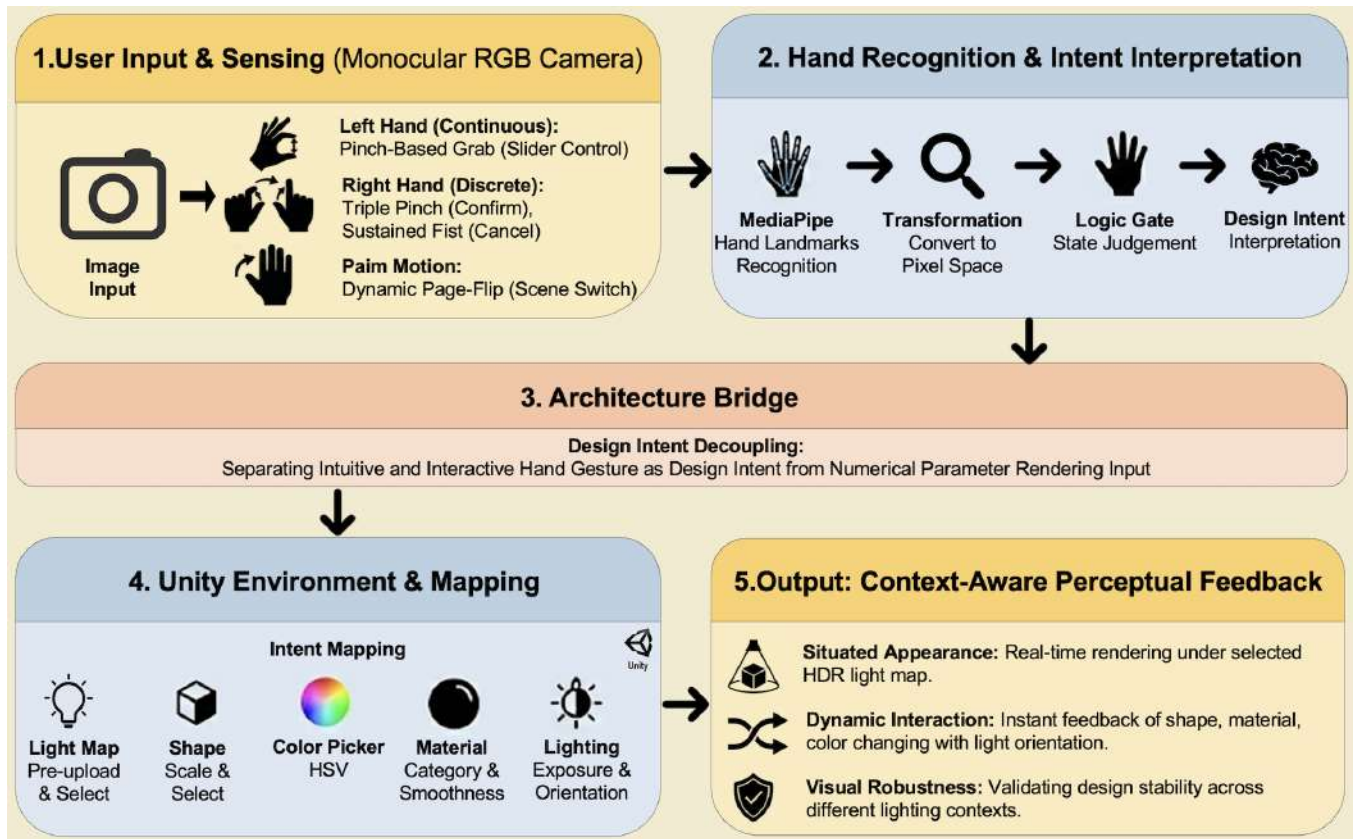


Figure 2: System architecture illustrating the decoupled intent layer between gesture sensing and perceptual rendering. Raw hand-tracking data is translated into design intents before being mapped to visual attributes in Unity, ensuring stable interaction despite noisy mid-air input.

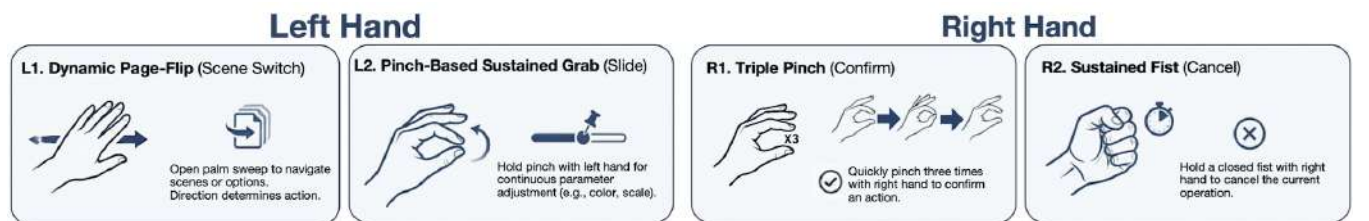


Figure 3: The core gesture vocabulary. L1) Dynamic Page-Flip for scene switching. L2) Pinch-Based Sustained Grab for continuous adjustment. R1) Triple Pinch for discrete confirmation. R2) Sustained Fist for cancellation.

allow designers to stress-test the stability of appearance judgments under variation. This separation foregrounds lighting not as a static backdrop, but as an active design material that shapes how form, material, and colour are interpreted in context.

The system is not intended to replace professional rendering tools or support final-stage photorealistic production. Instead, it targets early-stage exploration, where designers often rely on empirical judgement due to the lack of perceptually grounded tools. By lowering technical and cognitive barriers, the demo enables users to directly engage with the interplay of shape, material, colour, and lighting without specialised rendering expertise.

In conclusion, this CHI demo presents a walk-up-and-use design environment in which lighting is treated as an active design material rather than a post-process. By translating established perceptual findings into immersive, embodied interaction, the system illustrates how perceptual knowledge can be embedded into interactive tools to support exploratory design thinking in XR.

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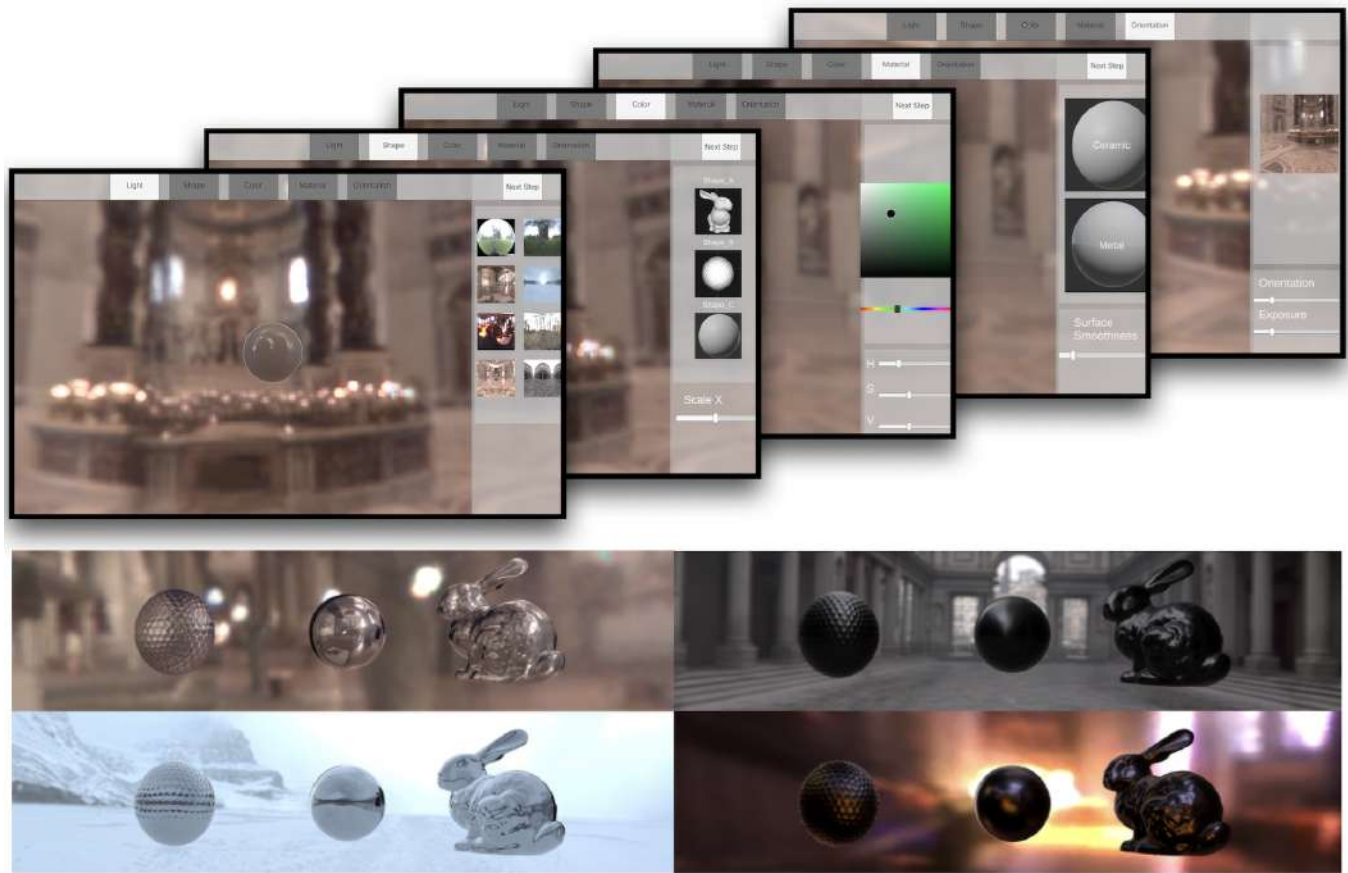


Figure 4: Top: User interface illustrating the five phases of the workflow. Bottom: Example rendering results under four different light maps [6], shown across three predefined shapes and two materials. Please note that users explore one shape at a time. Here multiple shapes are shown for illustrative comparison only.

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